PHY 205, General Physics II, Spring 2026

Instructor:	Nelson Coates (he, him, his)
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Class Time/Days/Location:	205-A: 9:15 am – 10:10 am, MWF, DB 232 205-B: 12:30 pm – 1:25 pm, MWF, BC 209
Course Pre- and Co-requisites:	PHY 204, MTH 202 (Pre, with a grade of C- or higher) PHY 275 (Co)

Some Planned Office Times:

10:20 am – 11:20 am, MWF
1:30 pm – 2:30 pm, MWF

I'm usually available many other times during the day and also after 9 pm (e.g. on MS Teams)

You are also very welcome to set up a time to meet outside of the hours listed or to just stop by my office unannounced. If my office door is closed when you stop by, please knock to check if I'm in!

Through the development of Maxwell's equations, we are going to explore the classical physics and applications of Electricity and Magnetism. Once we have motivated and learned to apply these equations, we will then examine how they contributed to the emergence of the drastic changes in the physical interpretations of reality that resulted around the turn of the 20th century.

Classroom Environment

There are many ways that you may want to document and reflect on your learning in this course, but your letter grade at the end of the semester is based only on your documented evidence of expertise with our course learning objectives. The most effective way for everyone in this course to receive the letter grade that they would like is by leveraging the strengths of the group and so you are also encouraged to collaborate with any of your work. To have the opportunity to draw on the abilities and goals of *all* of our members, we will prioritize our relationships, assume positive intent, be mindful of our contextual roles and responsibilities, and work to minimize authority in our teaching and learning. Physics needs all of us, and we all need physics – thank you for contributing!

Moodle

Copies of the course materials such as the syllabus, major assignment handouts, etc. may be found on Moodle. You are responsible for regularly checking Moodle for announcements and course updates.

Textbook

We will be using Volume 2 of the free OpenStax textbook *University Physics*.

Course AI Policy

I'm interested to learn how AI might support your learning in this course. Please let me know when you use AI tools, and also feel free to reach out to me and/or each other for suggestions to try with using AI.

Attendance

If you need to miss a class, please let me know beforehand so that I can make any adjustments to the class activities that would need your participation and make sure you are able to effectively stay up to date with the info and materials that you may miss.

Video/Audio Recording

Please do not record (audio or video) in this class except by permission, or in accordance with ADA accommodations. Any recordings made in connection with an accommodation are for that student's personal academic use only and may not be distributed to any other individual. Any virtual classroom sessions will be recorded and made available on the course Moodle site. These recordings of this course are for your personal academic use only. *Students may not distribute or share any images or recordings from the course without the permission of the instructor.*

Philosophy of Feedback

My goal is to engage with you in a mutually respectful exchange of feedback about our teaching and learning throughout this semester. In particular, I am eager to know about any barriers to your full participation in this course that I might be able to help remove.

In order to implement our learning cycles of *guidance* → *practice* → *feedback* & *revision* with our learning objectives, the main feedback that you will receive on your work for this class will be directly linked to our learning objectives and will have a goal of being *informative & motivational* for your learning. Therefore, all the feedback on any of your work this semester up until the last day of our class can be used to revise your work to demonstrate your progressive learning with our course content. Revisions for your work on our learning objectives should include a statement that *explains what you learned* from your earlier attempts (e.g. from the feedback) and *how that learning increases your expertise* with that objective.

Course Learning Objectives and Assessments

You will be provided with a list of our course learning objectives. You are encouraged to demonstrate your expertise with these objectives in ways that are more meaningful to you, but I will offer at least three specific options for you to demonstrate each objective. Your final course grade will be based on your documented evidence of expertise with the course learning objectives, and the evidence for each objective will be assessed as having either **(S)**uccessfully demonstrated your expertise, or having demonstrated your **(G)**rowing expertise. To be assessable between these two levels, your work for any objective also needs to *identify & justify* in your work where you have used each of the following three problem-solving approaches (epistemic frames) used by physicists: Conceptual Physics **(CP)**, Algorithmic Physics **(AP)**, and Algorithmic Math **(AM)**. Use of the fourth epistemic frame - Conceptual Math **(CM)** will catch you flair and is encouraged but not required. Templates to help you identify your progress using these epistemic frames with our objectives that are created in both MS Excel and Google Sheets formats will be posted on Moodle.

Reassessments

Unless you are confident that your learning and logic are self-evident, please include with revisions of your work a brief (e.g. single sentence) description of: What you learned the previous version of the work on that objective (e.g. from feedback) that increased your demonstrated expertise.

Opt-in to 'vouching' (peer assessments)

After **(S)**uccessfully demonstrating expertise with an objective, anyone can *opt-in* to being a 'voucher' for that objective. Being a voucher for an objective means that you have the ability to formally assess at the level required by our course whether our classmates' work for that objective indicates a **(G)**rowing expertise, or **(S)**uccessfully demonstrates their expertise.

I will keep a link on our Moodle page to a list of classmates (with their email addresses) who are vouchers for our objectives so that people can reach out to request assessment on their work. If you get in touch with a voucher who is unavailable or if you don't get a timely response, please reach out to me (or a different voucher) and I will do my best to prioritize your assessment.

If you'd like to become a voucher for any of your eligible objectives, please let me know so that we can have a brief conversation to talk about common misunderstandings for that objective before I add your name to the list of vouchers that is linked on Moodle.

After a voucher has provided feedback and determined that someone's work for an objective shows either their **(G)**rowing expertise or has **(S)**uccessfully demonstrated their expertise, the voucher can let me know in an email that says something like: *'I vouch that* [classmate] *has demonstrated their growing expertise in objective* [number]*'* or *'I vouch that* [classmate] *has successfully demonstrated their expertise with objective* [number]*'.*

Course Letter Grade : Your final letter grade for the course will be matched to the criteria below.

A **(S)**uccessfully demonstrated expertise in all 31 of our objectives.

A- **(S)**uccessfully demonstrated expertise in 28 objectives &
(G)rowing expertise in the remaining 3 objectives.

B+ **(S)**uccessfully demonstrated expertise in 27 objectives &
(G)rowing expertise in 3 of the remaining 4 objectives.

B **(S)**uccessfully demonstrated expertise in 25 objectives &
(G)rowing expertise in 4 of the remaining 6 objectives.

B- **(S)**uccessfully demonstrated expertise in 24 objectives &
(G)rowing expertise in 4 of the remaining 7 objectives.

C+ **(S)**uccessfully demonstrated expertise in 23 objectives &
(G)rowing expertise in 5 of the remaining 8 objectives.

C **(S)**uccessfully demonstrated expertise in 23 objectives &
(G)rowing expertise in 5 of the remaining 8 objectives.

C- **(S)**uccessfully demonstrated expertise in 20 objectives &
(G)rowing expertise in 6 of the remaining 11 objectives.

Due Dates and Deadlines

Although the only truly significant deadline for this class is the end of the semester, deadlines exist in the real world often for the purpose of creating a more efficient workflow of tasks that transfer between multiple people (e.g. feedback and revision). There is usually some flexibility with deadlines though and if you need some extra time, you can get in touch with whoever set the deadline to ask for an extension. If you ask for extensions often however, people might start wondering if you are ok. This class works in a similar manner - it's more efficient to provide feedback in batches and the feedback can be better crafted for your learning when there is less time pressure, so if producing work for feedback becomes a barrier for your participation in this course then I will probably get in touch to ask if you are ok and if you would like any support with engaging with our class.

PHY 205 – General Physics II, Spring 2025 - *Tentative Schedule*

Week	Topics, Readings
(1) Jan 12	Chapter 5 : Electric Charge, Force, and Field
(2) Jan 19	Chapter 5 : Electric Charge, Force, and Field
(3) Jan 26	Chapter 6 : Gauss's Law
(4) Feb 2	Chapter 6/7 : Gauss's Law and Electric Potential
(5) Feb 9	Chapter 7/8 : Electric Potential and Electrostatic Energy and Capacitors
(6) Feb 16	Chapter 8/9 : Electrostatic Energy, Capacitors, and Electrical Current
(7) Feb 23	Chapter 9 : Electrical Current and Resistance
(8) Mar 2	Spring Vacation
(9) Mar 9	Chapter 10 : Direct Current (DC) Circuits
(10) Mar 16	Chapter 11 : Magnetic Forces and Fields
(11) Mar 23	Chapter 12 : Sources of Magnetic Fields (Easter Vacation Friday)
(12) Mar 30	Chapter 13 : Electromagnetic Induction (Easter Vacation Monday)
(13) Apr 6	Chapter 15 : Alternating Current (AC) Circuits
(14) Apr 13	Chapter 16 : Maxwell's Equations and Electromagnetic Waves
(15) Apr 20	Review and Work on Objectives

ONCE UPON A TIME, THERE WAS A UNIVERSE. We are not sure about how it started or whether there is a reason. We don't know, for example, if spacetime is ordered or disordered at the smallest scales, which are dominated by the weirdness of quantum mechanics. We are pretty sure that during the first sliver of a trillionth of a second it expanded very rapidly so that for the most part it looked the same in every direction and looked the same from every position. It was sameness everywhere. Except that particles started to blip out of nothing due to random fluctuations caused by quantum effects, maybe in spacetime, we are still not super sure about that. Then again, we are not super sure about this, either: for some reason those particles formed more matter than antimatter. That process, which formed a particle type called baryons, is called baryogenesis. From there those baryons started to form structures, and from those structures stars formed. Then the stars got old and some of them died in super epic, rather fabulous fashion. They exploded into supernovae, making heavy elements like carbon and oxygen in the process. Those elements went on to be the basis for all life on Earth. Earth is a planet, one of the structures that formed around stars from the leftovers of supernovae. Eventually, a smaller type of structure that we call life formed on Earth. Some of the life-forms that evolved were relatively hairless apes that use a variety of methods of communication. There are about 7 billion of these apes, with various levels of eumelanin and pheomelanin in their skin and hair, giving them a range of colors. The apes also have a lot of different hair textures. Some of the ones with less eumelanin have for a long time now been cruel to the ones with more, some of whom we know as "Black people." We know why this is although we don't fully understand the why, but it might be due to laziness or because they are jealous of our boogie. But despite this, Black lives come from the same baryogenesis, the same supernovae, and the same structure formation. No matter what the lowest-eumelanin people say, Black Lives Are Star Stuff and Black Lives Matter-all of them.

-Chanda Prescod-Weinstein, *The Disordered Cosmos*

Academic Integrity Statement: The University of Portland is a diverse academic community of learners and scholars who are dedicated to freely sharing ideas and engaging in respectful discussion of those ideas to discover truth. Such pursuits require each person, whether student or faculty, to present truthfully our own ideas and give credit to others for the ideas that they generate. Thus, cheating on exams, copying another student's assignment, including homework, or using the work of others without proper citation are some examples of violating academic integrity.

Especially for written and oral assignments, students have an ethical responsibility to properly cite the authors of any books, articles, or other sources that they use. Students should expect to submit assignments to Turnitin, a database that ensures assignments are original work of the student submitting. Each discipline has guidelines for how to give appropriate credit, and instructors will communicate the specific guidelines for their discipline. The Clark Library also maintains a webpage that provides citation guidelines at https://libguides.up.edu/cite.

The misuse of AI to shortcut course learning outcomes will be treated as a violation of academic integrity comparable to plagiarism or cheating. Faculty are responsible for including a written "Course AI Policy" in their syllabi that clearly states what they consider appropriate and inappropriate uses of AI in the context of their courses. Students are responsible for using AI in ways that do not detract from the established learning outcomes of the course. All members of the scholarly community are responsible for demonstrating sound judgment in discerning when and how to utilize AI in their work, upholding standards of citation, originality, and integrity.

Assessment Disclosure Statement: Student work products for this course may be used by the University for educational quality assurance purposes. For reasons of confidentiality, such examples will not include student names.

Accessibility Statement: The University of Portland strives to make its courses and services fully accessible to all students. Students are encouraged to discuss with their instructors what might be most helpful in enabling them to meet the learning goals of the course. Students who experience a disability are encouraged to use the services of the Office for Accessible Education Services (AES), located in the Shepard Academic Resource Center (503-943-8985). If you have an AES Accommodation Plan, you should meet with your instructor to discuss how to implement your plan in this class. Requests for alternate location for exams and/or extended exam time should, where possible, be made two weeks in advance of an exam, and must be made at least one week in advance of an exam. Also, if applicable, you should meet with your instructor to discuss emergency medical information or how best to ensure your safe evacuation from the building in case of fire or other emergency. All information that students provide regarding disability or accommodation is confidential. All students are responsible for completing the required coursework and are held to the same evaluation standards specified in the course syllabus.

Mental Health Statement: Anyone can experience problems with their mental health that interfere with academic experiences and negatively impact daily life. If you or someone you know experiences mental health challenges at UP, please contact the University of Portland Counseling Center (https://www.up.edu/counseling/) in the upper level of Orrico Hall (down the hill from Franz Hall and near Mehling Hall) at 503-943-7134 or hcc@up.edu. Their services are free and confidential. In addition, mental health consultation and support is available through the Pilot Helpline by calling 503-943-7134 and pressing 3. All UP students also have access to teletherapy through BetterMynd. The University of Portland Campus Safety Department (503-943-4444) also has personnel trained to respond sensitively to mental health emergencies at all hours. Remember that getting help is a smart and courageous thing to do – for yourself, for those you care about, and for those who care about you. For more information on health and wellness resources at UP go to www.linktr.ee/wellnessUP .

Non-Violence Statement: The University of Portland is committed to fostering a safe and respectful community free from all forms of violence. Violence of any kind, and in particular acts of power-based personal violence, are inconsistent with our mission. Together, all UP community members must take a stand against violence. Learn more about what interpersonal violence looks like, campus and community resources, UP's prevention strategy, and what we as individuals can do to assist on the Green Dot website, www.up.edu/greendot. Further information and reporting options may be found on the Title IX website, www.up.edu/titleix.

Statement on Ethics of Information: The University of Portland is a community dedicated to the investigation and discovery of processes for thinking ethically and encouraging the development of ethical reasoning in the formation of the whole person. Using information ethically, as an element in open and honest scholarly endeavors, involves moral reasoning to determine the right way to access, create, distribute, and employ information, including: considerations of intellectual property rights, fair use, information bias, censorship, and privacy. More information can be found in the Clark Library's guide to the Ethical Use of Information at libguides.up.edu/ethicaluse.

The Learning Commons: Students may receive academic assistance through Learning Commons tutoring services and workshops. The Co-Pilot peer tutoring program provides students with opportunities to work with other students to get help in writing, math, group projects, and many other courses. Schedule an appointment to meet with a Co-Pilot (tutor) by visiting the Learning Commons website (www.up.edu/learningcommons). Students can also meet with a Co-Pilot during drop-in hours. Check the Learning Commons website or stop by the Learning Commons in BC 163 to learn more about their services. Co-Pilots are a wonderful support along your academic journey.